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Semester – V

Deep Learning

Project 3

Title: Comparison of Deep Learning Architectures

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Project 3

Comparison of Deep Learning Architectures

1. Introduction

Deep Learning is a part of Machine Learning that uses neural networks with multiple layers to learn patterns from data. Different types of neural networks are designed to work with different kinds of data. This project focuses on three important Deep Learning architectures: **Feedforward Neural Networks (FNNs)**, **Convolutional Neural Networks (CNNs)**, and **Recurrent Neural Networks (RNNs)**.

The purpose of this project is to understand the basic architecture of these networks, identify their suitable applications, and compare them based on their structure, input data, and real-world uses.

Task 1: Basic Architecture of FNN, CNN, and RNN

1.1 Feedforward Neural Network (FNN)

A Feedforward Neural Network is one of the basic types of neural networks. In this network, information moves in only one direction, from the input layer to the output layer.

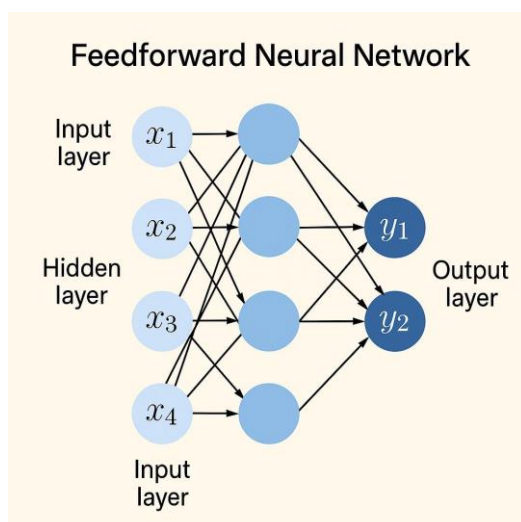
Main Components:

- **Input Layer:** Receives the input features.
- **Hidden Layers:** Process the input and learn patterns.
- **Output Layer:** Produces the final result.

Flow:

Input → Hidden Layers → Output

FNNs are mainly suitable for data where there is no important spatial or sequential relationship between the inputs.



1.2 Convolutional Neural Network (CNN)

A Convolutional Neural Network is mainly designed for processing **images and visual data**. CNNs can identify important features such as edges, shapes, textures, and objects.

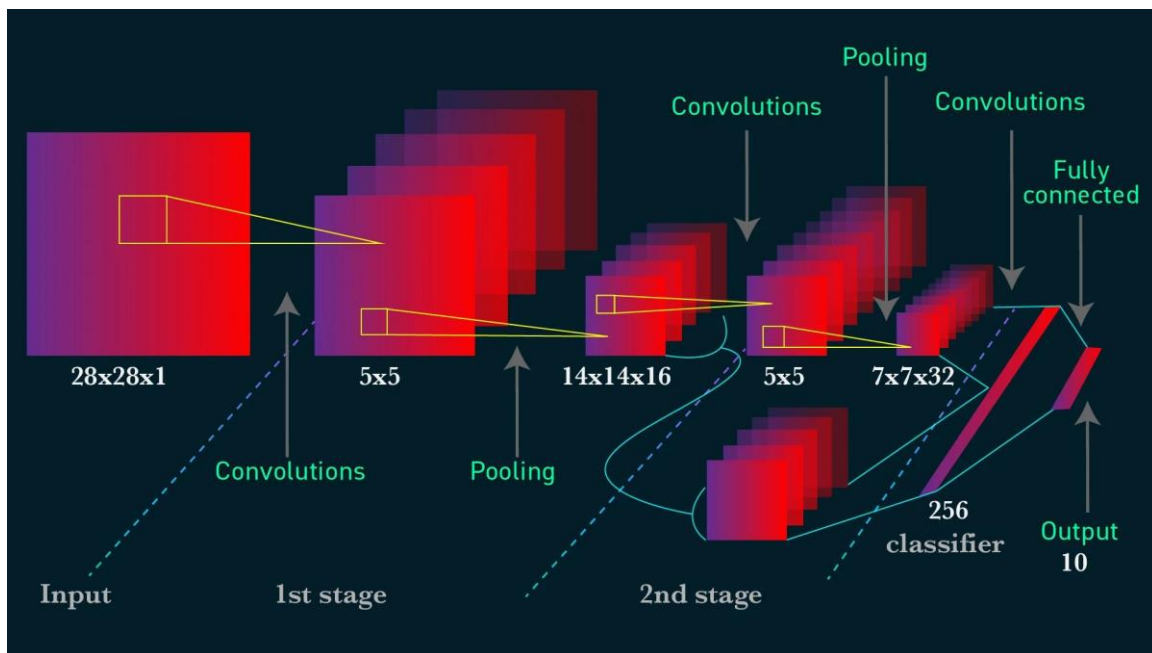
Main Components:

- **Convolution Layer:** Extracts important features from the image.
- **Pooling Layer:** Reduces the size of the feature maps.
- **Fully Connected Layer:** Uses the extracted features for classification.
- **Output Layer:** Gives the final prediction.

Flow:

Input Image → Convolution → Pooling → Fully Connected Layer → Output

CNNs are widely used when the input contains spatial information.



1.3 Recurrent Neural Network (RNN)

A Recurrent Neural Network is designed to work with **sequential data**. Unlike FNNs, RNNs can use information from previous inputs while processing the current input.

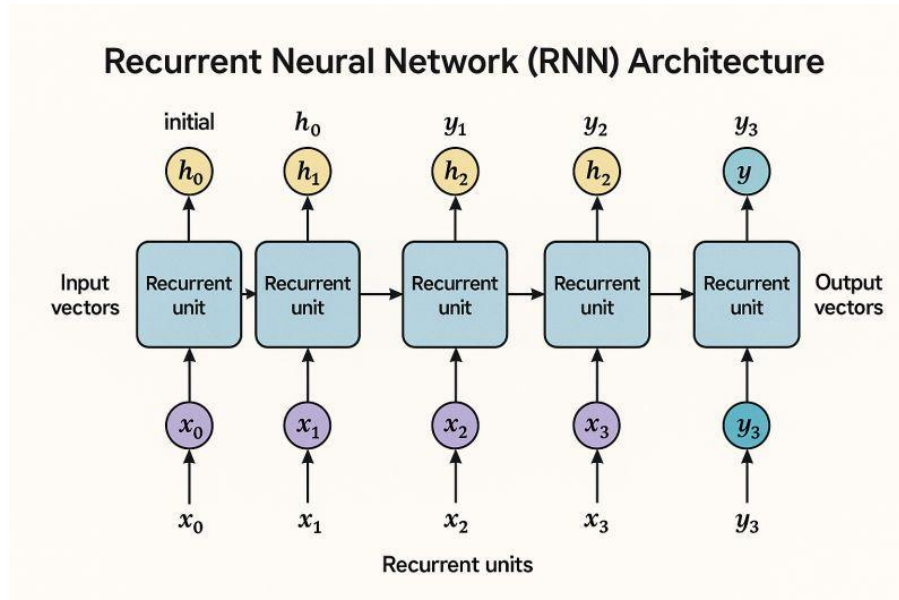
Main Components:

- **Input:** Receives data one step at a time.
- **Hidden State:** Stores information from previous steps.
- **Output:** Produces the result based on the current input and previous information.

Flow:

Previous Information + Current Input $\rightarrow$ RNN $\rightarrow$ Output

RNNs are useful for data where the order of information is important, such as text, speech, and time-series data.



Task 2: Suitable Applications

2.1 Applications of FNN

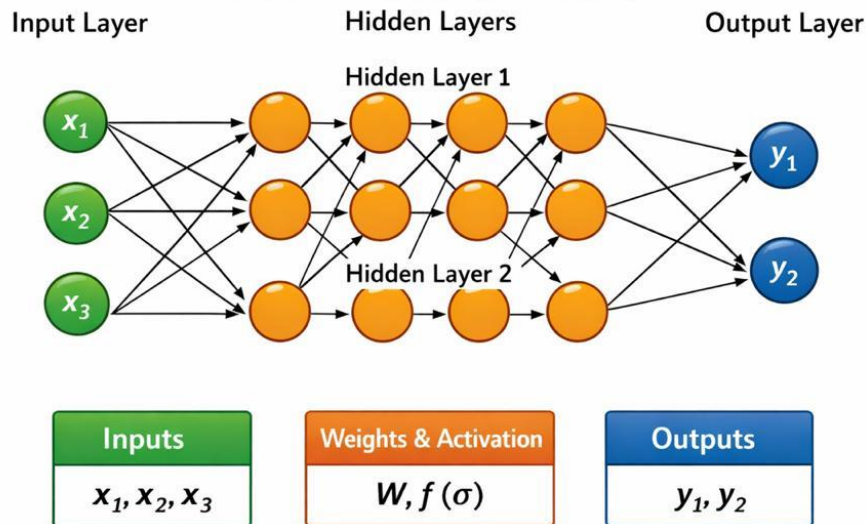
FNNs can be used for:

- Student performance prediction
- Customer churn prediction
- Loan approval prediction
- Sales prediction
- Fraud detection

Example:

An FNN can predict whether a customer may leave a company using information such as age, monthly spending, and subscription period.

Feedforward Neural Network



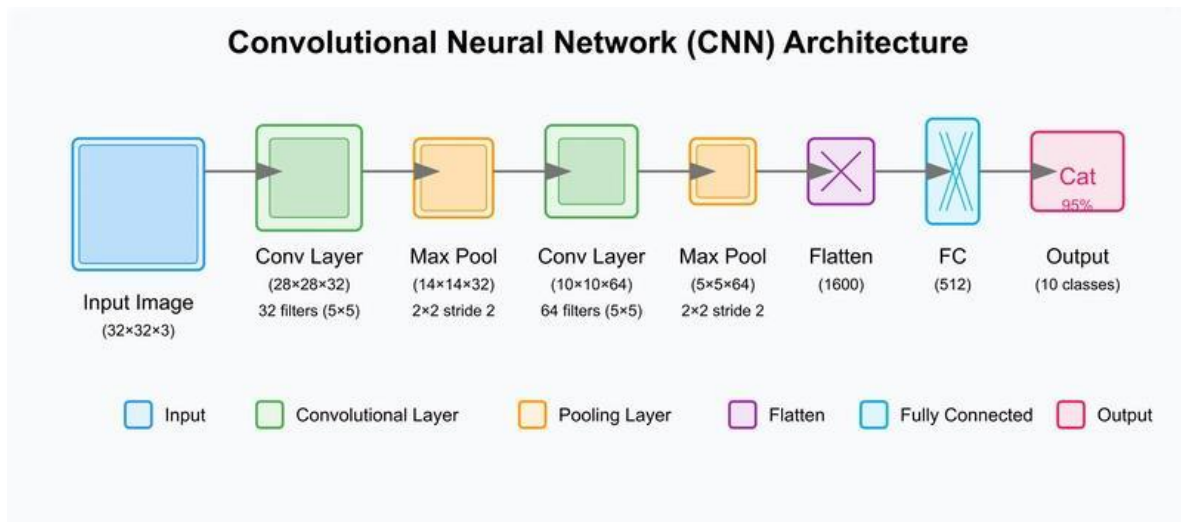
2.2 Applications of CNN

CNNs are commonly used for:

- Image classification
- Face recognition
- Object detection
- Medical image analysis
- Handwritten digit recognition

Example:

A CNN can identify whether an image contains a **cat or dog** by learning features such as edges, shapes, and textures.



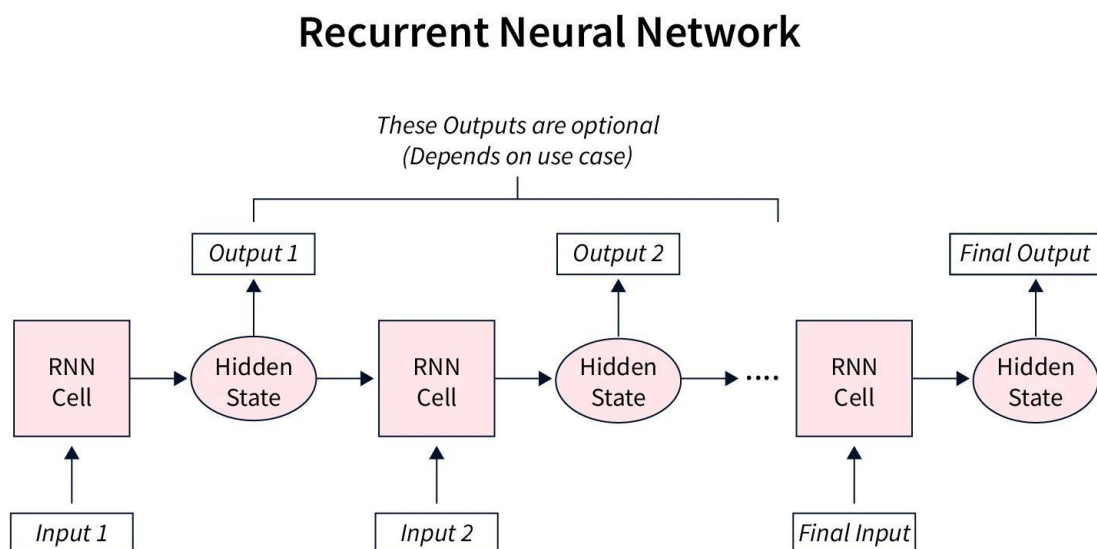
2.3 Applications of RNN

RNNs are commonly used for:

- Speech recognition
- Sentiment analysis
- Text generation
- Language translation
- Time-series prediction

Example:

An RNN can analyze a customer review and determine whether the sentiment is **positive**, **negative**, or **neutral**.



Task 3: Comparison of the Three Architectures

Aspect	Feedforward Neural Network (FNN)	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Basic Structure	Neurons arranged in fully connected layers; signal moves in a single forward direction	Layers of convolution and pooling filters followed by fully connected layers	Neurons with feedback loops; the hidden state is passed from one step to the next
Direction of Data Flow	One-directional (input to output only)	One-directional, but spatial filters slide over the input	Cyclical / sequential; output of a step feeds back as input to the next step
Type of Input Data	Fixed-size numeric or tabular data (structured data)	Grid-like data such as images, video frames	Sequential data such as text, speech, or time-series
Memory of Past Inputs	None	None (each input is processed independently)	Yes, retains context from earlier steps through hidden state
Key Mechanism	Weighted sums and activation functions	Convolution and pooling operations for feature extraction	Recurrent connections that carry information across time
Typical Applications	Prediction, classification of tabular data (e.g. loan approval)	Image classification, object detection, facial recognition	Language modelling, speech recognition, stock price prediction
Strength	Simple and fast for structured data	Excellent at capturing spatial patterns	Excellent at capturing temporal/sequential patterns
Limitation	Cannot handle spatial or sequential patterns well	Not designed for sequential/time-dependent data	Can be slow to train; struggles with very long sequences (addressed by LSTM/GRU)
Everyday Example (from this report)	Email spam detection from structured features / loan approval prediction	Classifying a photo as 'Cat' or 'Dog'	Predicting the next word while typing on a keyboard
Well-Known Real-World Systems	Loan prediction	Cat/dog classification	Sentiment analysis

Conclusion

Feedforward Neural Networks, Convolutional Neural Networks, and Recurrent Neural Networks are important Deep Learning architectures designed for different types of problems. **FNNs** are suitable for structured data and general prediction tasks. **CNNs** are mainly used for images and other visual data because they can identify spatial features. **RNNs** are suitable for sequential data because they can use information from previous inputs.

Therefore, the choice of architecture depends on the **type and structure of the input data and the problem being solved**. Understanding these differences helps in selecting the appropriate Deep Learning architecture for real-world applications.